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C.F.G.S. DAM
Cross-Platform Application Development

Markup languages languages and information management systems (LMS) Data Bases	
Duración:17 /09/2026 – 24/11/2026	***Organización: Grupos 3/4 personas***



Challenge 1.2
My first Application

Every company starts with an idea. Some are born in a garage, others in a coffee shop, and many simply emerge from people who observe a problem and think, "What if we did it differently?"

In this challenge, you will be the entrepreneurs. You will need to come up with an original business idea, define the products or services it will offer, and consider how it will create value for its customers. There are no limits: you could create a tech company, an educational platform, a specialty store, an app to improve sustainability, or any idea you are passionate about.

But a good idea needs a solid foundation to become a reality. That is why you will work on two fundamental aspects of application development:

The website will serve as your business's calling card. On it, you will showcase who you are, what you offer, and why your proposal stands out.

The database. You will need to analyze what information your business needs to store: customers, products, orders, reservations, users, suppliers, or any other relevant entity.



Objectives and Competencies

Technical competencies and Learning Outcomes (RAs):

LMS

RA1. Recognize the characteristics of markup languages by analyzing and interpreting code snippets.

RA2. Use markup languages to transmit information over the Web, analyzing document structure and identifying their elements.

BDA

RA 6. Design normalized relational models by interpreting Entity-Relationship diagrams.

Transversal competencies

See attached document on transversal competencies.

1. Tasks to be performed

- **Web Page (LMS)**

  In the LMS course, the task is to create a website consisting of a homepage that links to at least three other pages. All pages must have a return option to the homepage. Website navigation must be intuitive.

  The content of the different pages must be consistent with the selected theme and the format must be uniform.

  The files that make up the website should be separated into different folders (img, css, audio, web for example).

  The use of flexbox or Bootstrap is mandatory for organizing the content of each page. Each webpage must have at least four sections (header, menu, content, and footer). Additionally, at least one page must have three columns for its content. The website will adapt to changes in screen size.

  All CSS code must go in an external stylesheet.

  Mandatory elements on the website:

  - Lists: ordered and/or unordered
  - Tables: at least 2 with headers and/or row and/or column joins
  - Floating frame (iframe)
  - Images
  - Image map
  - All types of links (internal, local, remote)
  - Audio
  - Video (link to online videos; do not add your own files)
  - Advanced form with all types of elements, properties, and restrictions
  - A menu using links. One of the menu options will be a dropdown with more choices(an example will be shown).

  Use of required styles in:

  - Text
  - • Lists
  - • Tables (each with different styles, for example: striped even/odd rows, highlighted headers or footers, borders, hover effects by row, etc.) - an example will be shown
  - • Links (normal, visited, active, and hover effects).
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  - • Forms (change the default appearance of elements, focus, highlight the border, etc.)
  - • Documents must be well-formed and will be validated on the website https://validator.w3.org/.

  The organization and clarity of the HTML and CSS code will also be valued (correct tabulation, use of meaningful CSS identifiers and selectors, etc.).

  If any of these requirements are not met, the challenge grade will automatically be a 0.

- **Data Bases (BDA)**

  Furthermore, for the BDA course, as part of the company project you are developing, you must design the database that will support the information displayed and managed via the corporate website.

  Before beginning the technical design, it will be necessary to analyze your company's information needs: what data will be stored, who will use it, how the data elements relate to one another, and what rules must be followed to ensure the integrity and consistency of the information.

  The result must be a well-structured database proposal capable of storing and managing all the information necessary for the company's operations.

  To do this, you must:

  1. Requirements analysis. Provide a detailed description of the information the database must store. This section will serve as the **statement of requirements** or task (*enunciado*) upon which the rest of the design will be built.
  2. Entity-Relationship Model Design. The model must faithfully represent the company's operations and allow for the management of all information identified during the analysis. The diagram must be created digitally.
  3. Transformation to the Relational Model. Based on the previous E-R diagram, the relational model will be required.
  4. Database Normalization. Starting from the previous relational model, normalize it to 3NF.

- **Presentación:**

  On the last day of the challenge, a presentation of the developed website and the designed database will be given. Key aspects of the site should be highlighted (such as compliance with all requirements, additional features, etc.). The team may present any aspects they deem appropriate, covering not only the final result but also the process of developing the challenge.

  All group members must participate equally in the presentation.

  The presentation will be evaluated based on the criteria outlined in the transversal competencies rubric.

  At the end of the presentation, the instructor may ask any team member questions they deem appropriate, even regarding a section that the specific member did not present or develop during the challenge; therefore, all team members must possess a comprehensive understanding of the entire project. Todos los integrantes del grupo deben participar en la presentación equitativamente.

- **Deliverables (in Moodle)**

  LMS

  - Zip file with all the files that make up the website (one per team).
  - PowerPoint (or similar) used in the presentation (one per team).

  BDA

  - Zip file containing the statement, the E/R model, the conversion to the relational model and its subsequent normalization (you can submit it in a single document or separately).
  - PowerPoint (or similar) used in the presentation (one per team).

2. Evaluation criteria

- **TECHNICAL COMPETENCIES**

  Each group will be evaluated based on the technical rubric for each module

  As this challenge involves collaborative work, every team member must have a thorough understanding of all the phases undertaken and the technical details of the proposed solution. Each member must be able to explain, justify, and defend the project and the decisions made during its execution.

- **TRANSVERSAL COMPETENCIES**

  - Teacher-student assessment 90%
  - Student assessment - student (in teams) 5%
  - Self-assessment 5%

Grades will be individual.

See attached rubric for cross-curricular skills.

3. Resources

Resources available in ***Retos 1DAM/1DAMi-retos intermodulares*** Moodle course.

4. Timing

Recommended timing 1DAM/1DAMi	
Steps / Phases	**Week**
1. Activate the device 2. Plantear el reto 3. Identify and connect with the challenge 4. Set parameters	16-17 september
5. Obtain and organize the information	17 september - 30 october
6. Generate alternatives 7. Submit proposal 8. Select proposal 9. Plan actions	2 - 13 november
10. Implement actions and individual evaluation 11. Present results and initial evaluation 12. Evaluate results and feedback	16 - 20 november